



SYSTEM ANALYSIS & DESIGN

Management Information System Department
3rd Year - 2nd Semester

Lec #4

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- The approval committee may authorize the system analyst to prepare a more detailed about the proposed IS project.
- Feasibility analysis:
 - Guides the organization in determining whether to proceed with the project.
 - Identifies the important risks associated with the project.
 - Assesses three areas:
 - 1 Technical feasibility,
 - 2 Economic feasibility, and
 - 3 Organizational feasibility.
- A feasibility study deliverable that is submitted to the approval committee at the end of project initiation.

- 1** Technical feasibility: Can we build it?
 - a** **Familiarity with application:**
 - Less familiarity generates more risk.
 - b** **Familiarity with technology:**
 - Less familiarity generates more risk.
 - c** **Project size:**
 - Large projects have more risk.
 - d** **Compatibility:**
 - The harder it is to integrate the system with the company's existing technology, the higher the risk will be.

2 Economic Feasibility: Should we build it?

- Development costs.
- Annual operating costs.
- Annual benefits (cost savings and/or increased revenues) .
- Intangible benefits and costs.

3 Organizational Feasibility: If we build it, Will they come?

- Project champion(s).
- Senior management.
- Users.
- Other stakeholders.
- Is the project strategically aligned with the business?

1- Technical Feasibility

- The extent to which the system can be successfully designed, developed, and installed by the IT group.
- Risks:
 - 1 Familiarity with the application:
 - When analysts/developers are unfamiliar with the business application area, they have a greater chance of misunderstanding the users or missing opportunities for improvement.
 - This risk increases dramatically when the users themselves are less familiar with an application
 - 2 Familiarity with the technology:
 - When a system will use technology that has not been used before within the organization.
 - This risk increases dramatically when the technology itself is new.

■ Risks:

3 Project size:

■ Measured by:

- a Length of project completion time.
- b Number of the development team.
- c Number of features in the system.

4 Compatibility:

- New technology and applications need to be able to integrate with the existing environment for many reasons.
 - a Rely on data from existing systems.
 - b Use the existing communications infrastructure.

- It is also called a cost-benefit analysis.
- Refers to the ability of a project or business to generate enough revenue to cover its costs and provide a reasonable return on investment.
- Steps:
 - a) Identify costs and benefits.
 - b) Assign values to costs and benefits.
 - c) Determine cash flow.
 - d) Assign project economic value.

a) Identify Costs and Benefits

- The system analyst's first task is to identify the kinds of costs and benefits the system.
- List the tangible costs and benefits for the project. Include both one-time and recurring costs.

Development Costs	Operational Costs
Development Team Salaries	Software Upgrades
Consultant Fees	Software Licensing Fees
Development Training	Hardware Repairs
Hardware and Software	Hardware Upgrades
Vendor Installation	Operational Team Salaries
Office Space and Equipment	Communications Charges
Data Conversion Costs	User Training

a) Identify Costs and Benefits ...

Tangible Benefits	Intangible Benefits
Increased Sales	Increased Market Share
Reductions in Staff	Increased Brand Recognition
Reductions in Inventory	Higher Quality Products
Reductions in IT Costs	Improved Customer Service
Better Supplier Prices	Better Supplier Relations

b) Assign Values to Costs and Benefits & c) Determine Cash Flow

- System analyst works with business users and IT professionals to create **numbers** for each of the costs and benefits.
- Even intangibles should be valued if at all possible.
- The value of the project must be measured over a period of time (usually three to five years).
- Cash flows (inflows and outflows) are estimated over some future period.

c) Determine Cash Flow ...

	2005	2006	2007	2008	2009	Total
Benefits						
Increased Sales		500,000	530,000	561,800	595,508	2,187,308
Reduction in Customer Complaint Calls ^a		70,000	70,000	70,000	70,000	280,000
Reduced Inventory Costs		68,000	68,000	68,000	68,000	272,000
Total Benefits^b		638,000	668,000	699,800	733,508	2,739,308
Development Costs						
2 Servers @ \$125,000	250,000	0	0	0	0	250,000
Printer	100,000	0	0	0	0	100,000
Software Licenses	34,825	0	0	0	0	34,825
Server Software	10,945	0	0	0	0	10,945
Development Labor	1,236,525	0	0	0	0	1,236,525
Total Development Costs	1,632,295	0	0	0	0	1,632,295
Operational Costs						
Hardware		50,000	50,000	50,000	50,000	200,000
Software		20,000	20,000	20,000	20,000	80,000
Operational Labor		115,000	119,600	124,384	129,359	488,343
Total Operational Costs		185,000	189,600	194,384	199,359	768,343
Total Costs	1,632,295	185,000	189,600	194,384	199,359	2,400,638
Total Benefits – Total Costs	(1,632,295)	453,000	478,400	505,416	534,149	338,670
Cumulative Net Cash Flow	(1,632,295)	(1,179,295)	(700,895)	(195,479)	338,670	
Return on Investment (ROI)	14.1%	(338,670/2,400,638)				
Break-Even Point	3.37 years	(costs are fully recovered in year 4: 534,149 – 338,670) / 534,149 = .37)				

^a Customer service values are based on reduced costs of handling customer complaint phone calls.

^b An important yet intangible benefit will be the ability to offer services that our competitors currently offer.

- Evaluate the project's expected returns in comparison to its costs using:

1 Return on Investment (ROI):

- A calculation that measures the average rate of return earned on the money invested in the project.
- A high ROI suggests that the project's benefits far outweigh the project's cost.

$$\text{ROI} = \frac{\text{Total Benefits} - \text{Total Costs}}{\text{Total Costs}}$$
$$= (2739308 - 2400638) / 2400638 = 14.1\%$$

- Evaluate the project's expected returns in comparison to its costs using:
 - Break Even Point (BEP) (a.k.a The Payback Method):
 - Defined as the number of years it takes a firm to recover its original investment in the project from net cash flows.

$$\text{BEP} = \begin{array}{c} \text{Number of} \\ \text{years of} \\ \text{negative} \\ \text{cash flow} \end{array} + \frac{\text{That year's Net Cash Flow} - \text{That year's Cumulative Cash Flow}}{\text{That year's Net Cash Flow}}$$

$$= 3 + (534149 - 338670) / 534149 = 3.37 \text{ years}$$

d) Assign Project Economic Value ...

■ Cumulative net cash flow

=Cumulative of previous year+Net benefit of current
year

$$(\text{year 1}) = -100.000 + 35.000 = -65.000$$

$$(\text{year 2}) = -65.000 + 38.000 = -27.000$$

$$(\text{year 3}) = -27.000 + 41.000 = 14.000$$

	Year 0	Year 1	Year 2	Year 3	Total
Total Benefits		45,000	50,000	57,000	152,000
Total Costs	100,000	10,000	12,000	16,000	138,000
Net Benefits (Total Benefits – Total Costs)	(100,000)	35,000	38,000	41,000	14,000
Cumulative Net Cash Flow	(100,000)	(65,000)	(27,000)	14,000	

3- Organizational Feasibility

- How well the system ultimately will be accepted by its users and incorporated into the ongoing operations of the organization.
- Ways to assess:
 - 1 Understand how well the goals of the project align with business objectives.
 - 2 Conduct a stakeholder analysis.
- A stakeholder is a person, group, or organization that can affect (or will be affected by) a new system.

Important Stakeholders for Organizational Feasibility

	Role
Champion	A champion: • Initiates the project • Promotes the project • Allocates his or her time to project • Provides resources
Organizational Management	Organizational Managers: • Know about the project • Budget enough money for the project • Encourage users to accept and use the system
System Users	Users: • Make decisions that influence the project • Perform hands-on activities for the project • Ultimately determine whether the project is successful by using or not using the system

Task 3: Feasibility Study

- The task is to be carried out by teams.
- In this assignment, you are going to prepare a feasibility analysis for your own choice IS project.
- This assignment has three parts. They are:
 - 1 Technical feasibility.
 - 2 Economic feasibility.
 - 3 Organizational feasibility.